CARS24 · TAKE-HOME ASSIGNMENT

MSME Financial Health Card

Explainable credit decisions for new-to-credit businesses, built on consent-based alternate data

50,000 MSMEs scored | 0–100 score + 6 pillars | Real-time REST API | SHAP explanations

Deepesh Sharma · November 2026

THE PROBLEM

No credit history ≠ not creditworthy

42%

of the 50,000 businesses are **new-to-credit**: no loan, no bureau score

73%

of new-to-credit businesses are **not high-risk** on alternate data

78%

of existing borrowers are not high-risk: **almost the same**

A bureau-only lender rejects all ~21,000 of them. GST, UPI, bank and payroll data show that most of them can repay.

THE SOLUTION

The solution in one picture

01

Collect

With consent: bank data via AA, GST, UPI, EPFO payroll, invoices

02

Engineer

33 raw inputs plus 11 analyst ratios, peer and seasonal features

03

Predict

Default risk, health score, 6 pillar scores and anomaly checks

04

Decide

Risk band, loan limit, tenure, rate and conditions

05

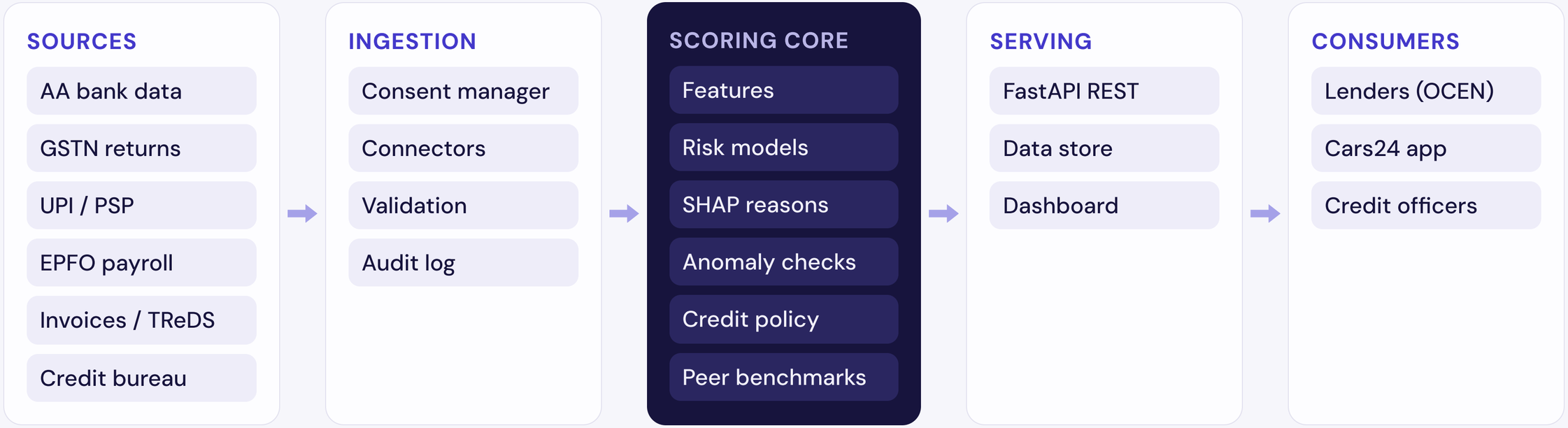
Explain

SHAP reasons, peer ranking and what-if actions

Output: one Financial Health Card per business, in under a second, served through a REST API and a dashboard.

DELIVERABLE 1 · SOLUTION ARCHITECTURE

End-to-end architecture



Design rule: one scoring core is shared by training, the API and the dashboard, so features are computed the same way everywhere.

DELIVERABLE 1 · DATA INGESTION PIPELINE

Data ingestion: consent first

1. Borrower applies for credit in the Cars24 app
2. We send a consent request through an Account Aggregator: purpose, data types, date range
3. Borrower approves in their AA app; we receive a signed consent artefact
4. Fetch: bank data from AA, GST via a GSP, payroll from EPFO, UPI from the PSP
5. Validate, merge and store; every step goes to the audit log
6. Score in under a second: Health Card and loan offer go back to the app

Built in the demo

- Mock consent artefact (POST /consents)
- One mock connector per data source
- Pydantic range checks on every field
- SQLite: records, cards, consents, audit

Production upgrade

- AA consent manager with expiry and revocation
- Real connectors behind the same interface
- Kafka topic per source, retries, dead-letter queue
- Postgres, encrypted raw store, feature store

DELIVERABLES 1 & 3 · AI/ML WORKFLOW

AI/ML workflow

50k MSMEs → EDA → Features → 70/15/15 split → **Ridge + LightGBM** → Test-set checks → Batch scoring

LightGBM, monotonic

Main model. Constraints on 14 features: filing GST late can never raise a score.

Ridge baseline

Scorecard-style check. It matches LightGBM because the data is near-linear.

Three targets

Default probability (as log-odds), the 0–100 health score and the 6 pillar scores.

TreeSHAP

Exact per-feature contributions, shown as score points and decline reason codes.

Anomaly detection

8 cross-source rules plus an Isolation Forest that flags the most unusual 2%.

Policy layer

Kept apart from the model, as lenders do: band → eligibility → limit → conditions.

DELIVERABLE 2 · SAMPLE CARD FOR MSME0000001

The Financial Health Card

FINANCIAL HEALTH SCORE

77.8 / 100

Grade B | Default risk 7.8% · Medium

CREDIT DECISION

Approve with conditions

₹3.05 L · 24 months · 18% p.a.

EMI ₹15,227 · Product: used commercial-vehicle finance from Cars24

Business Stability
73.9 A
Beats 76% of MSMEs

Cash Flow
69.6 B
Beats 45% of MSMEs

Revenue Consistency
74.7 A
Beats 85% of MSMEs

Payment Behaviour
81.7 C
Beats 39% of MSMEs

Business Growth
54.5 A
Beats 88% of MSMEs

Compliance
86.0 B
Beats 59% of MSMEs

RISK INDICATORS

- High · customers take 34 days to pay
- Watch · lean-month surplus ₹13,253
- Watch · staff attrition 23%
- OK · no fraud flags

DELIVERABLE 3 · MODEL QUALITY

Model results on 7,500 unseen businesses

0.98
AUC for spotting high-risk businesses

83%
risk band exactly right; every miss is one step away

92%
eligibility decisions match the supplied ones

0.98
AUC for new-to-credit firms: as good as for existing borrowers

Model	PD R^2	Band accuracy	Eligibility	High-risk AUC
Ridge baseline (scorecard-style)	0.918	83.2%	92.4%	0.979
LightGBM + monotonic constraints	0.912	82.9%	92.4%	0.977
Ceiling: the supplied PD itself	1.000	100%	100%	1.000

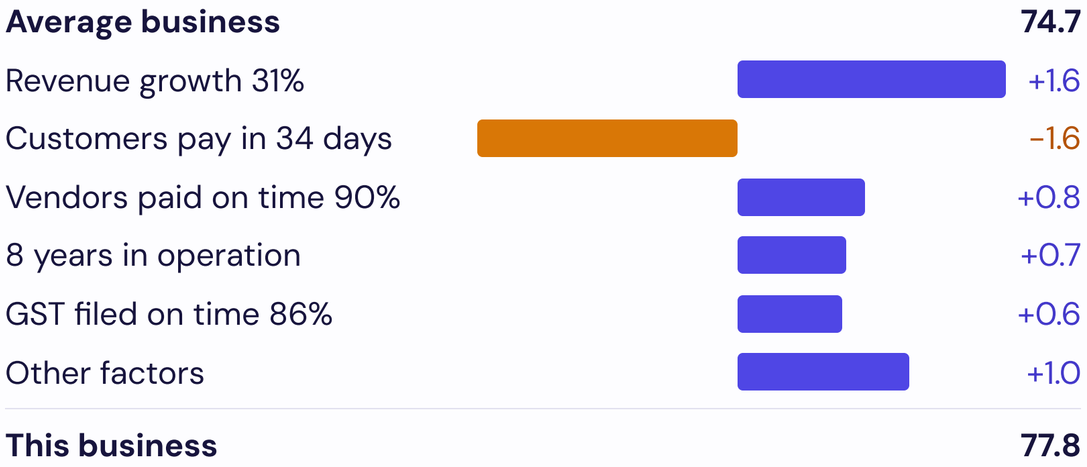
Finding: the simple baseline matches LightGBM. The supplied scores are near-linear formulas plus random noise; even an oracle model scores the same, so we are at the noise ceiling.

DELIVERABLE 3 · EXPLAINABILITY & ANOMALIES

Every decision explains itself

Why MSME0000001 scores 77.8

SHAP: each factor's push, in score points



Fraud checks across sources

Tested by planting fake cases in clean data

Planted pattern	Caught
Circular transactions	100%
Inflated GST sales	100%
Hidden debt burden	100%
Ghost payroll	40%
False alarms, clean firms	2.5%

Ghost payroll needs monthly EPFO detail to catch reliably: a known gap.

DELIVERABLE 5 · API DESIGN

REST APIs for every job in the brief

Data ingestion

`POST /consents`	Create consent
`POST /msmes`	Full record
`POST /msmes/{id}/sources/{src}`	Push one source
`POST /msmes/{id}/fetch`	Pull all sources

Score generation

`POST /score`	Real-time, stateless
`POST /msmes/{id}/score`	Score and save

Health Card & credit

`GET /msmes/{id}/health-card`	Full card
`GET /msmes/{id}/credit-recommendation`	Offer

Dashboard integration

`GET /dashboard/msmes/{id}`	One business
`GET /dashboard/portfolio`	KPIs
`GET /dashboard/msmes`	Filtered list

All paths sit under **/api/v1** · interactive OpenAPI docs at **/docs** · 16 automated tests cover the full flow

DELIVERABLE 6 · INTEGRATION STRATEGY

Integration with India’s credit rails

Account Aggregator

Cars24 registers as an FIU. Consented, encrypted bank statements give credits, balances, overdraft use and EMIs.

GSTN

Through a GST Suvidha Provider with OTP consent: GSTR-1 sales, GSTR-3B filing dates, GSTR-2B purchases.

OCEN

Cars24 acts as the Loan Service Provider. The Health Card JSON travels with the loan request; lenders price the offer.

ULI

Each ULI data service becomes one more connector. New sources add features; the API contract stays the same.

Banking APIs & UPI

A lender’s own statement APIs and PSP settlement data: faster than AA for existing customers.

EPFO & bureau

EPFO gives headcount, wage bill and on-time PF. The bureau is optional: blank for NTC by design.

In code: one Connector interface per source. The demo uses mocks; production swaps in real connectors without touching the scoring core.

DELIVERABLE 1 · TECHNOLOGY STACK

Tech stack and scalability

Layer	Technology
Language	Python 3.12
Data	pandas, NumPy, Parquet
ML	LightGBM, scikit-learn, SHAP
API	FastAPI, Pydantic, Uvicorn
Dashboard	Streamlit, Plotly
Storage	SQLite → Postgres
Testing	pytest (16 tests)
Deploy	Render, Docker

0.4 s
full card with SHAP for 8 models

45 ms
score only, for real-time journeys

12 s
to score all 50,000 businesses

350 MB
memory: fits Render's free tier

At scale: stateless API behind a load balancer · Kafka per data source · Redis feature store · MLflow registry with champion–challenger

WHERE LLMS AND NEWER AI FIT

Use LLMs around the score, not instead of it

Read messy documents

Vision LLMs pull figures from PDF bank statements, invoices and GST filings when AA data isn't available.

Understand transactions

An LLM tags each UPI or bank narration as rent, supplier, salary or EMI: richer features for the model.

Explain in plain words

An LLM turns SHAP reason codes into a short note for the officer or owner, in Hindi or English.

Underwriter copilot

An agent calls our APIs (card, peers, what-if) to answer "why was this declined?" with real numbers.

Tabular foundation models

Pre-trained tabular models such as TabPFN as a challenger to LightGBM, useful for small new segments.

Graph and sequence models

Graph ML on the GST buyer–supplier network to find fraud rings; transformers on monthly cash flows.

Guardrail: the loan decision stays with the explainable model. LLMs read, tag and explain; they don't approve loans, because a lender must be able to justify every decision.

CARS24 ANGLE & NEXT STEPS

Why this fits Cars24

Products matched to the business

Logistics & fleets → used commercial-vehicle finance

Traders & retailers → inventory and working-capital lines

Construction & healthcare → equipment finance

As an OCEN Loan Service Provider, Cars24 can bring credit to dealers and fleet owners that banks can't assess today.

Next steps with real data

- Train on real repayment: 90+ days past due
- Monthly AA and GST history for true trend charts
- Champion–challenger: Ridge vs LightGBM, live
- Drift monitoring (PSI) and monthly retraining

Dashboard: msme-health-dashboard.onrender.com

API: msme-health-api-yx6e.onrender.com/docs

Code: github.com/deepesh463/msme-health-card